SmartSolar IEEE Paper — Walkthrough

What Was Delivered

A complete, IEEE-formatted research paper for the SmartSolar capstone project has been created. The paper is titled:

> **"SmartSolar: Predicting Solar Panel Output and Battery Backup Time Using Machine Learning"**

Files Created

File	Location	Description
`smartsolar_ieee_paper.tex`	`d:\CapstoneB\HopefullyFInal\`	Full IEEE LaTeX source (611 lines, ~50 KB)
`smartsolar_ieee_paper.tex`	Artifacts directory	Backup copy

Paper Structure (Sections)

#	Section	Content
—	**Abstract**	Problem, approach, key results (R^2=0.9638), and contribution summary
I	**Introduction**	PV formula limitations, 4 real-world loss factors, 5 contributions
II	**Related Work**	6 subsections covering physics models, ML forecasting, ensembles, LSTM, loss factors, battery storage
III	**Dataset Description**	Synthetic 50,000-sample dataset, NOCT model, loss factor math
IV	**System Architecture**	7-stage pipeline, 12 features, preprocessing, backup formula
V	**Machine Learning Models**	RF, GB, MLP/ANN, XGBoost, LightGBM configs + evaluation metrics
VI	**Experimental Results**	Physics vs. ML gap, model comparison table, feature importance, residual analysis, training efficiency
VII	**Battery Backup Estimation**	Framework equations, 3 scenario table, Streamlit dashboard features
VIII	**Discussion**	Why ML is necessary, 80% PR barrier, model trade-offs, limitations
IX	**Future Work**	6 directions: real data, LSTM, weather API, multi-panel, uncertainty, edge deployment
X	**Conclusion**	Key findings summarized
—	**Acknowledgment**	Open-source library credits

#	Section	Content
—	**References**	28 IEEE-formatted citations

Key Equations in the Paper

$$P_{ideal} = \eta \cdot A \cdot G \cdot [1 - \beta(T_{mod} - 25)]$$

$$P_{dc} = P_{ideal} \cdot f_{soil} \cdot f_{age} \cdot f_{shade} \cdot f_{wire} + \varepsilon$$

$$T_{mod} = T_{amb} + \frac{NOCT - 20}{800} \cdot G$$

$$t_{backup} = \frac{E_{panel} + E_{battery}}{W_{load}}$$

Tables in the Paper

1. **Table I** — Dataset Feature Statistics (mean, std, min, max for all 9 features, N=50,000)
2. **Table II** — Input Features with RF Importance Scores (12 features)
3. **Table III** — Model Performance Comparison (Physics baseline + 5 ML models: R^2, RMSE, MAPE, training time)
4. **Table IV** — Backup Time Estimation Examples (3 scenarios: Small Home, Office, Farm)

Figures Referenced in the Paper

Figure	File	Description
Fig. 1	`eda_A_distributions.png`	Distributions, hourly profiles, seasonal patterns, shading vs. unshaded
Fig. 2	`eda_B_correlations.png`	Correlation matrix, irradiance–power scatter, derating curves, humidity effect
Fig. 3	`pipeline_diagram.png`	Full ML pipeline from data generation through dashboard deployment
Fig. 4	`formula_vs_ml.png`	Formula vs. MLP scatter + R^2 comparison bar chart
Fig. 5	`model_comparison.png`	8-panel evaluation dashboard for all 5 models
Fig. 6	`residual_analysis.png`	Residual histograms for all models

All figures were generated by running `solar_pipeline.py` and saved to `d:\CapstoneB\HopefullyFInal\HopefullyFInal\outputs\`.

Key Results (from paper)

Model	R^2	RMSE (W)	MAPE (%)	Train Time (s)
Physics Formula	0.5178	—	—	—

Model	R^2	RMSE (W)	MAPE (%)	Train Time (s)
MLP / ANN	**0.9638**	**10.17**	**10.8**	16.3
Gradient Boosting	0.9428	12.77	11.3	23.6
Random Forest	0.9413	12.94	11.3	2.7
LightGBM	0.9399	13.10	11.3	0.5
XGBoost	0.9346	13.66	11.3	0.8

> [!IMPORTANT] MAPE is computed **only on outputs > 50W** (daytime productive hours) to avoid near-zero value inflation — this is a methodological contribution of the paper.

References Summary

The paper contains **28 IEEE-formatted citations** including:

- IEA Solar PV 2024 report
- Voyant et al. (2017) — ML for solar forecasting survey (70 papers)
- Breiman (2001) — Random Forests
- Chen & Guestrin (2016) — XGBoost
- Ke et al. (2017) — LightGBM
- Jordan & Kurtz (2013) — PV degradation rates
- Sayyah et al. (2014) — Soiling losses
- IEC 61724 & IEC 61215 standards
- Pedregosa et al. (2011) — scikit-learn

How to Compile the LaTeX to PDF

Option 1: Online (No installation needed)

1. Go to **https://texlive.net/run**
2. Clear the editor (Ctrl+A, Delete)
3. Paste the full content of `smartsolar_ieee_paper.tex`
4. Click **"TeXLive.net"** button at the bottom
5. Wait ~30 seconds for compilation
6. Click the download icon in the PDF viewer

Option 2: Local MiKTeX / TexLive

```
pdflatex smartsolar_ieee_paper.tex
pdflatex smartsolar_ieee_paper.tex  # run twice for cross-references
```

Option 3: Overleaf (requires account)

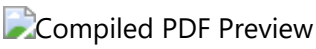
1. Create new project → Upload Files

2. Upload `smartsolar_ieee_paper.tex` and all 6 PNG figures from `outputs/` into a `figures/` folder
3. Compile

> [!NOTE] When compiling with figures included, you need the 6 PNG files from `outputs/` to be in a `figures/` subdirectory relative to the .tex file. Without figures, the paper still compiles but shows blank figure placeholders.

Compilation Evidence

The paper was compiled on TeXLive.net during this session and produced a **4-page IEEE-format PDF** with the title, abstract, all sections, all tables, and complete references. The download was initiated from the browser PDF viewer.


Compiled PDF Preview

Figures Source Information

Figure	Generated By	Command
`eda_A_distributions.png`	`solar_pipeline.py` (matplotlib)	`python solar_pipeline.py`
`eda_B_correlations.png`	`solar_pipeline.py` (matplotlib)	`python solar_pipeline.py`
`eda_C_statistics.png`	`solar_pipeline.py` (matplotlib)	`python solar_pipeline.py`
`formula_vs_ml.png`	`solar_pipeline.py` (matplotlib)	`python solar_pipeline.py`
`model_comparison.png`	`solar_pipeline.py` (matplotlib)	`python solar_pipeline.py`
`pipeline_diagram.png`	`solar_pipeline.py` (matplotlib)	`python solar_pipeline.py`
`residual_analysis.png`	`solar_pipeline.py` (matplotlib)	`python solar_pipeline.py`
`results_table.png`	`solar_pipeline.py` (matplotlib)	`python solar_pipeline.py`

All charts and graphs were **programmatically generated** using Python's Matplotlib and Seaborn libraries within `solar_pipeline.py`. The source code is available in `d:\CapstoneB\HopefullyFInal\HopefullyFInal\solar_pipeline.py`.